

Q1. What is cybersecurity?

A. Cybersecurity is the practice of protecting systems, networks, applications, and data from cyber threats and unauthorized access.

Q2. CIA triad?

A. The CIA Triad consists of Confidentiality, Integrity, and Availability, which are the core principles of information security.

Q3. What is encryption?

A. Encryption converts readable data into unreadable ciphertext using cryptographic algorithms to protect information.

Q4. Symmetric vs asymmetric encryption?

A. Symmetric encryption uses the same key for encryption and decryption, while asymmetric encryption uses public and private key pairs.

Q5. What is hashing?

A. Hashing is a one-way process that converts data into a fixed-length value used for integrity verification.

Q6. SSL/TLS?

A. SSL/TLS are cryptographic protocols that secure communication over networks through encryption and authentication.

Q7. What is HTTPS?

A. HTTPS is the secure version of HTTP that uses TLS encryption to protect data transmitted between clients and servers.

Q8. What is a firewall?

A. A firewall monitors and filters network traffic based on predefined security rules.

Q9. IDS vs IPS?

A. An IDS detects suspicious activity and generates alerts, while an IPS can automatically block malicious traffic.

Q10. What is phishing?

A. Phishing is a social engineering attack that tricks users into revealing sensitive information.

Q11. What is ransomware?

A. Ransomware is malware that encrypts data and demands payment for decryption.

Q12. What is malware?

A. Malware is malicious software designed to damage, disrupt, or gain unauthorized access to systems.

Q13. Zero-day exploit?

A. A zero-day exploit targets a software vulnerability before a patch or fix is available.

Q14. Social engineering?

A. Social engineering manipulates people into revealing confidential information or performing insecure actions.

Q15. What is MFA?

A. Multi-Factor Authentication requires multiple verification methods to confirm a user's identity.

Q16. What is a VPN?

A. A Virtual Private Network encrypts internet traffic and provides secure remote access.

Q17. DNS spoofing?

A. DNS spoofing redirects users to fraudulent websites by altering DNS responses.

Q18. SQL injection?

A. SQL injection is an attack that manipulates database queries through unsanitized user input.

Q19. What is XSS?

A. Cross-Site Scripting injects malicious scripts into web applications to target users.

Q20. What is CSRF?

A. Cross-Site Request Forgery tricks authenticated users into performing unintended actions.

Q21. Privilege escalation?

A. Privilege escalation occurs when attackers gain higher access rights than originally authorized.

Q22. Penetration testing?

A. Penetration testing is the authorized simulation of cyberattacks to identify security weaknesses.

Q23. Vulnerability scanning?

A. Vulnerability scanning uses automated tools to detect known security flaws.

Q24. OWASP Top 10?

A. The OWASP Top 10 is a list of the most critical web application security risks.

Q25. What is SIEM?

A. Security Information and Event Management (SIEM) collects, analyzes, and correlates security events.

Q26. Incident response?

A. Incident response is the process of identifying, containing, investigating, and recovering from security incidents.

Q27. Digital forensics?

A. Digital forensics involves collecting and analyzing digital evidence during investigations.

Q28. Packet sniffing?

A. Packet sniffing captures and analyzes network traffic for monitoring or troubleshooting.

Q29. What is Wireshark?

A. Wireshark is a network protocol analyzer used to inspect and troubleshoot network traffic.

Q30. TCP handshake?

A. The TCP three-way handshake establishes a connection using SYN, SYN-ACK, and ACK messages.

Q31. What is DDoS?

A. A Distributed Denial-of-Service attack overwhelms a target with traffic from multiple systems.

Q32. What is a botnet?

A. A botnet is a network of compromised devices controlled by an attacker.

Q33. Authentication vs authorization?

A. Authentication verifies identity, while authorization determines access permissions.

Q34. What is Active Directory?

A. Active Directory is Microsoft's directory service for managing users, devices, and permissions.

Q35. Endpoint security?

A. Endpoint security protects devices such as laptops, desktops, and mobile devices from threats.

Q36. Network segmentation?

A. Network segmentation divides networks into smaller sections to improve security and limit attack spread.

Q37. Threat intelligence?

A. Threat intelligence provides information about current and emerging cyber threats.

Q38. Risk assessment?

A. Risk assessment identifies, analyzes, and prioritizes security risks to an organization.

Q39. What is compliance?

A. Compliance means adhering to legal, regulatory, and industry security requirements.

Q40. Security auditing?

A. Security auditing evaluates systems, policies, and controls to ensure effectiveness and compliance.

Q41. What would you do after a breach?

A. Contain the incident, preserve evidence, investigate root cause, remediate vulnerabilities, and restore operations.

Q42. Prioritize vulnerabilities?

A. Prioritize based on severity, exploitability, business impact, and asset criticality.

Q43. SOC operations?

A. Security Operations Center teams monitor, detect, investigate, and respond to security threats.

Q44. MITRE ATT&CK?

A. MITRE ATT&CK is a framework that catalogs attacker tactics, techniques, and procedures.

Q45. What is EDR?

A. Endpoint Detection and Response provides continuous monitoring and response capabilities for endpoints.

Q46. Describe a security incident

A. Explain detection, investigation, containment, eradication, recovery, and lessons learned.

Q47. Threat investigation example

A. Describe how logs, alerts, and threat intelligence were used to identify and mitigate a threat.

Q48. How do you stay updated?

A. Stay updated through security blogs, threat reports, certifications, conferences, and hands-on labs.

Q49. Biggest cyber threat today?

A. Common answers include ransomware, phishing, supply-chain attacks, and AI-powered cyber threats.

Q50. Questions for us?

A. Ask about security tools, team structure, incident response processes, training opportunities, and security challenges.